

Model practical Exam

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MLA0405

Deep Learning

Slot - D

12/1/2020

29) Neural Network Analysis for Multi-class Data
(Tanh Activation, 3 Hidden Neurons)

off
viva

31) Neural Network Analysis for Multi-class Data
(ReLU Activation, Learning Rate 0.03)

off
viva

23) Neural Network Analysis for Spiral Data (sigmoid
Activation)

off
viva

25) Neural Network Analysis for Circular Data (Tanh
Activation)

off
viva

Questions	Aim & Algorithm (20)	Program (45)	Execution (output) (25)	Viva (10)	Total marks (100)
1					
2					
3					
4					
Total marks					

23) Aim: To Implement a neural network using Sigmoid activation to classify spiral data.

Algorithm:

1. Generate spiral dataset
2. Split data into training and testing sets.
3. Initialize weights and biases.
4. Apply Sigmoid activation in the hidden and output layers.
5. Calculate error using loss function.
6. update weights using backpropagation.
7. Test the trained model and calculate accuracy.

Python Code:

```
import numpy as np
from sklearn.neural_network import
MLPClassifier
x = np.array([[0,0],[1,1],[1,0],[0,1]])
y = [0,1,1,0]
model = MLPClassifier(hidden_layer_sizes = (5,),
activation = 'logistic', max_iter = 1000)
model.fit(x,y)
Print("Predicted:"model.predict(x))
```

```
Print("Accuracy:"model.score(x,y))
```

Sample Input: x-data points, y-class labels

Sample output: predicted classes and accuracy

Result: Neural network classifies the data using Sigmoid activation.

25) Aim: To classify Circular data using a neural network with tanh activation.

Algorithm:

1. Generate circular data.
2. Initialize the neural network.
3. Apply tanh activation.
4. Train the network using backpropagation.
5. Predict the classes and calculate accuracy.

Python code:

```
import numpy as np
from sklearn.datasets import make_circles
from sklearn.neural_network import MLPClassifier
x, y = make_circles(200, noise=0.1, factor=0.5)
model = MLPClassifier(hidden_layer_sizes=(5, 2),
activation='tanh', max_iter=1000)
model.fit(x, y)
Print("Prediction:",
model.predict(x[:5]))
Print("Accuracy:", model.score(x, y))
Input: circular data points (x, y)
output: predicted class labels and accuracy.
Result: The neural network successfully classifies circular data
using tanh activation.
```

29) Aim: To classify multi-class data using a neural network with tanh activation and 3 hidden neurons.

Algorithm:

1. Generate multi-class data.
2. Create a neural network with 3 hidden neurons.
3. Use tanh activation.
4. Train the network.
5. Predict the classes and calculate accuracy.

Python code:

```
from sklearn.datasets import  
make_classification  
from sklearn.neural_network import  
MLPClassifier  
x, y = make_classification(n_samples=300, n_classes=3,  
n_informative=4, random_states=1)  
model = MLPClassifier(hidden_layer_sizes=(3, ),  
activation='tanh', max_iter=1000)
```

Print("Prediction:",

model.predict(x[:5]))

Print("Accuracy:", model.score(x, y))

Input: Multi-class dataset with 3 classes.

Output: predicted class labels and accuracy.

Result: The neural network successfully classified the multi-class data

using tanh activation with 3 hidden neurons.

31) Aim: To classify multi-class data using a neural network with ReLU activation
& learning rate 0.03

Algorithm

1. Generate multi-class data.

2. Create a neural network

3. Use ReLU activation & learning rate 0.03.

4. Train the network.

5. Predict classes and calculate accuracy.

Python code:

```
from sklearn.datasets import
```

```
make_classification
```

```
from sklearn.neural_network import
```

```
MLPClassifier
```

```
x, y = make_classification(n_samples=300, n_classes=3,
```

```
n_informative=4, random_states=(5, ),
```

```
activation='relu', learning_rate_init=0.03,
```

```
m.fit(x, y) max_iter=1000)
```

```
Print("Prediction:", m.predict(x[:5]))
```

```
Print("Accuracy:", m.score(x, y))
```

Input: Multi-class dataset with 3 classes

Output: predicted class labels & accuracy.

Result: The neural network successfully classified the multi-class data using ReLU activation & Learning rate 0.03